

MATUA, Russian Federation

WMO: 322070

Lat: 48.051N

Lon: 153.255E

Elev: 190

StdP: 14.60

Time Zone: 11.00 (E11)

Period: 86-95

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	12.7	14.9	2.7	6.4	15.5	5.0	7.2	16.7	42.6	21.2	39.1	26.3	14.0	270	0.876

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	9.6	65.5	61.4	61.6	58.1	58.4	55.2	62.9	64.2	58.7	60.7	55.9	57.7	17.1	240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
62.2	84.3	64.3	57.9	72.1	59.4	55.0	64.7	56.6	28.6	64.1	25.7	60.9	23.9	57.7	73.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 38.1	33.5	29.4	DB	8.5	70.1	3.3	4.9	6.1	73.7	4.2	76.6	2.3	79.3	-0.1	82.9	(4)
(5)			WB	7.1	65.6	2.9	4.4	5.0	68.8	3.2	71.3	1.6	73.8	-0.5	77.0	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
Temperatures, Degree-Days and Degree-Hours	DBAvg	36.9	23.7	24.4	27.4	32.0	37.2	41.8	47.4	52.1	49.5	43.7	34.5	28.4	
	DBStd	10.45	4.79	5.19	3.84	3.27	3.70	4.52	5.15	4.53	3.57	3.56	4.39	4.44	
	HDD50	4942	814	716	702	539	397	251	114	27	51	198	464	669	
	HDD65	10247	1279	1136	1167	989	861	695	546	399	466	661	914	1134	
	CDD50	171	0	0	0	0	2	5	34	93	35	2	0	0	
	CDD65	1	0	0	0	0	0	0	1	0	0	0	0	0	
	CDH74	1	0	0	0	0	0	0	1	0	0	0	0	0	
	CDH80	0	0	0	0	0	0	0	0	0	0	0	0	0	
Wind	WSAvg	12.8	15.7	16.0	16.8	15.6	11.6	8.5	8.1	8.9	10.5	13.0	14.3	14.9	
Precipitation	PrecAvg	37.3	2.2	1.8	2.1	2.5	2.8	2.8	3.9	4.1	4.4	4.0	3.7	3.0	
	PrecMax	45.3	6.0	3.5	3.8	4.6	4.3	4.0	6.5	11.1	6.9	6.1	5.9	6.1	
	PrecMin	28.6	0.9	0.7	1.1	0.7	0.9	0.8	1.2	0.5	2.2	2.2	1.7	1.4	
	PrecStd	4.3	1.1	0.7	0.6	0.8	0.8	0.9	1.3	2.0	1.2	1.0	1.2	1.3	
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	35.0	34.9	36.7	45.2	58.1	65.7	72.2	68.8	63.1	55.5	46.4	40.8	
		MCWB	33.5	33.3	35.7	39.8	48.3	55.7	67.5	64.4	59.6	51.9	45.0	39.5	
	2%	DB	33.6	33.5	33.9	39.5	49.4	57.5	66.0	65.4	59.7	52.3	44.4	38.2	
		MCWB	32.2	32.1	32.6	35.3	43.1	50.6	62.0	62.3	55.9	49.7	43.3	36.9	
	5%	DB	32.7	32.5	33.0	37.6	45.7	53.2	59.0	62.3	57.4	50.4	42.8	35.9	
		MCWB	31.4	31.2	31.7	35.6	41.1	47.8	55.6	59.7	54.7	48.3	41.8	34.3	
	10%	DB	31.4	31.8	32.2	36.2	42.8	49.2	54.6	59.4	55.2	49.0	41.2	34.1	
		MCWB	30.0	30.5	30.9	34.7	40.0	45.8	52.2	56.8	52.4	47.1	39.8	32.1	
	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	33.9	34.5	35.6	40.3	49.8	57.4	69.0	66.1	61.5	53.0	45.5	39.8
			MCDB	34.3	34.7	38.4	44.7	57.0	65.0	71.3	67.7	62.2	54.3	46.3	40.4
2%		WB	32.7	32.4	33.2	37.4	44.4	51.2	63.0	63.4	57.4	50.3	43.6	37.2	
		MCDB	33.3	32.9	33.5	38.0	48.1	55.6	64.9	64.9	59.1	51.5	44.1	38.1	
5%		WB	31.8	31.6	32.3	36.1	42.1	48.2	56.0	60.3	55.0	48.7	41.9	34.9	
		MCDB	32.6	32.3	32.6	37.0	44.6	52.4	58.4	61.4	56.5	50.1	42.6	35.9	
10%		WB	29.9	30.7	31.4	34.8	40.3	45.9	52.3	57.3	53.0	47.4	40.2	32.6	
		MCDB	31.3	31.7	32.0	35.8	42.2	48.4	54.1	58.5	54.4	48.8	41.1	33.7	
Mean Daily Temperature Range		5% DB	MDBR	5.5	5.5	5.5	5.0	7.5	9.0	8.8	9.6	8.1	7.1	6.1	5.8
			MCDBR	5.2	5.6	5.7	8.0	14.8	17.5	16.9	14.3	11.2	9.3	7.8	7.6
	MCWBR		5.6	5.4	5.9	6.5	9.8	11.9	13.3	11.2	9.4	7.9	7.8	8.2	
	5% WB	MCDBR	5.2	5.8	5.1	7.0	12.6	16.6	15.7	13.4	10.3	9.3	7.6	7.7	
		MCWBR	5.8	5.9	5.8	6.5	9.5	11.9	12.9	11.9	9.4	8.3	8.2	8.9	
Clear-Sky Solar Irradiance	taub	0.358	0.383	0.438	0.486	0.500	0.483	0.468	0.445	0.391	0.401	0.389	0.371		
	taud	2.126	2.081	1.977	1.903	1.936	2.060	2.175	2.245	2.360	2.156	2.091	2.054		
	Ebn at Noon	211	235	240	242	245	250	252	253	257	226	195	183		
	Edh at Noon	28	37	48	57	57	51	45	40	32	34	29	27		
All-Sky Solar Radiation	RadAvg	318	524	866	1134	1259	1326	1224	1095	1012	711	370	258		
	RadStd	22	48	70	131	169	126	135	90	88	50	30	20		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (1 neighbor)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page